

SIMATS ENGINEERING

ASSESSMENT TOOL II – DATA ANALYSIS PROJECT

Course: Data Handling and Visualization	Code: DSA0601	CO: CO3
Duration: 120 min	Total Marks: 100	Reg No : Name:

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Data Analysis Project

Weightage: 20%

Code of Conduct:

I ____R.Akshaya (192524118)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Akshaya. R

Criteria	Problem Understanding & Data Preparation 25%	Data Analysis & Interpretation 30%	Application of Analytical Techniques 20%	Analysis of Results 15%	Organization & Documentation 10%	Presentation & Clarity 5%	Total
Q1							
Q2							
Q3							

Q4

Q5

Data Analysis Project

Question 1 – Distribution Analysis Using ECDF and Q-Q Plot

Dataset: Employee_Salaries.csv

Assessment Task

Using the Employee_Salaries dataset, analyze the distribution of employee salaries. 1. Create an empirical cumulative distribution function (ECDF) for Salary.

```
library(ggplot2)
```

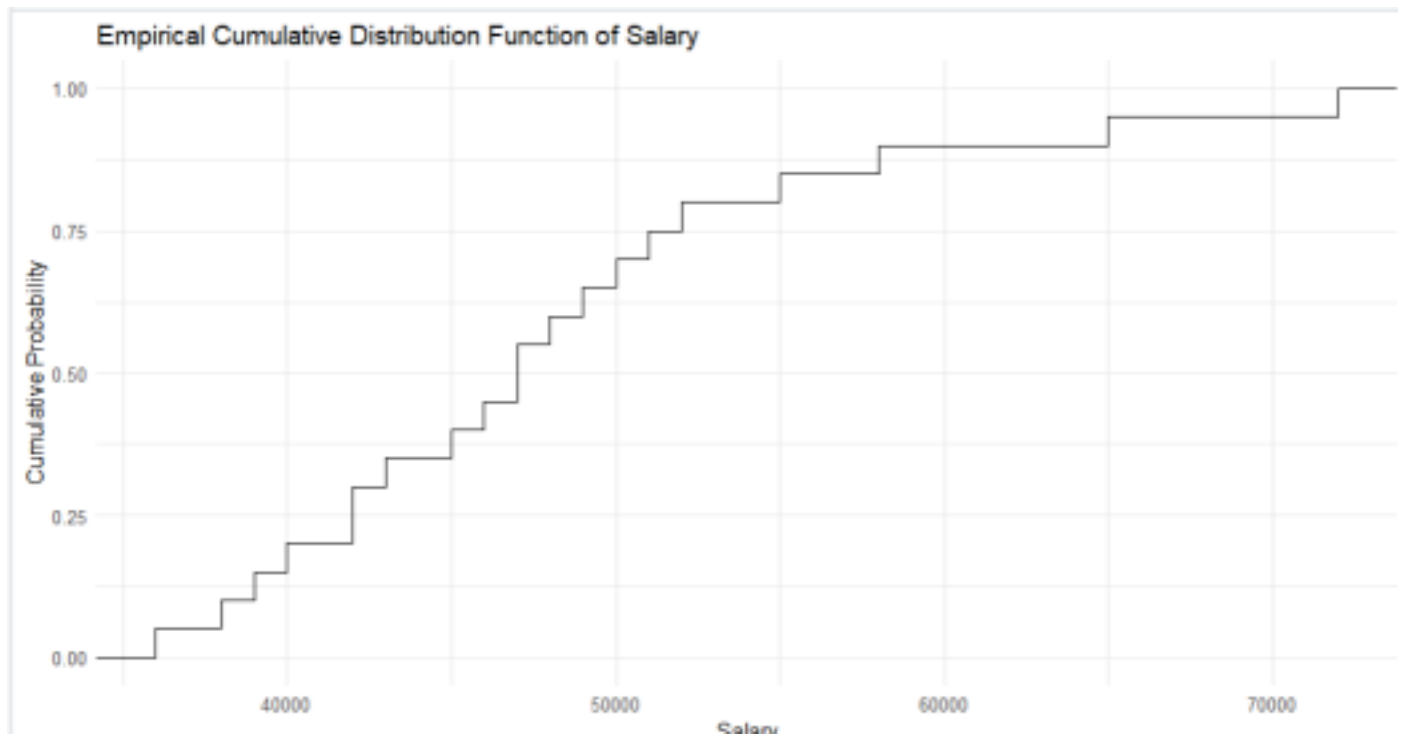
```
employee <- data.frame(  
  Employee_ID = paste0("E", sprintf("%03d", 1:20)),  
  Department = c(  
    rep("IT", 5),  
    rep("HR", 5),  
    rep("Finance", 5),  
    rep("Marketing", 5)  
  ),  
  Salary = c(  
    42000, 45000, 48000, 51000, 55000,  
    38000, 40000, 42000, 46000, 50000,  
    47000, 49000, 52000, 58000, 65000,  
    36000, 39000, 43000, 47000, 72000  
  )  
)
```

```
employee
```

	Employee_ID	Department	Salary
1	E001	IT	42000
2	E002	IT	45000
3	E003	IT	48000
4	E004	IT	51000
5	E005	IT	55000
6	E006	HR	38000
7	E007	HR	40000
8	E008	HR	42000
9	E009	HR	46000
10	E010	HR	50000
11	E011	Finance	47000
12	E012	Finance	49000
13	E013	Finance	52000
14	E014	Finance	58000
15	E015	Finance	65000
16	E016	Marketing	36000
17	E017	Marketing	39000
18	E018	Marketing	43000
19	E019	Marketing	47000
20	E020	Marketing	72000

2. Create a Q-Q plot to assess whether Salary is approximately normally distributed.

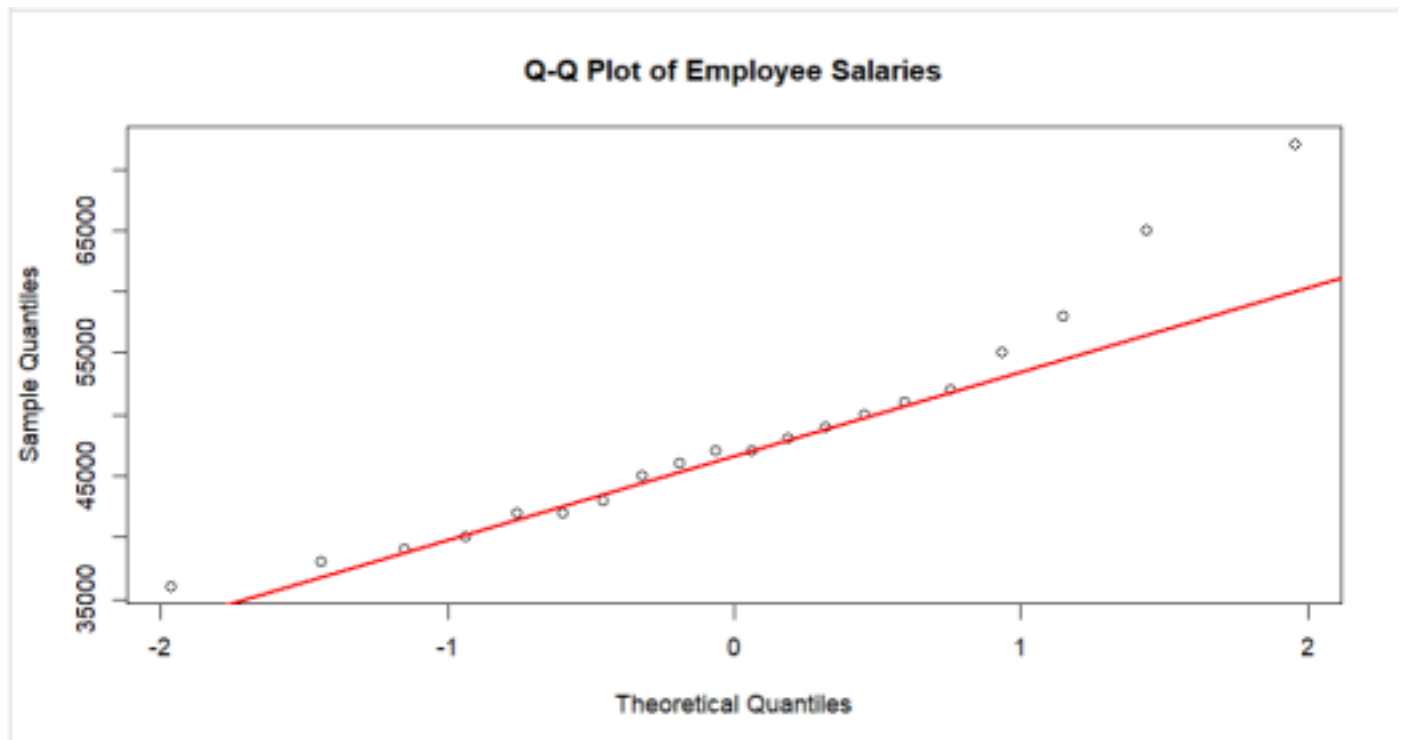
```
ggplot(employee, aes(x = Salary)) +
  stat_ecdf(geom = "step") +
  labs(
    title = "Empirical Cumulative Distribution Function of Salary",
    x = "Salary",
    y = "Cumulative Probability"
  ) +
  theme_minimal()
```



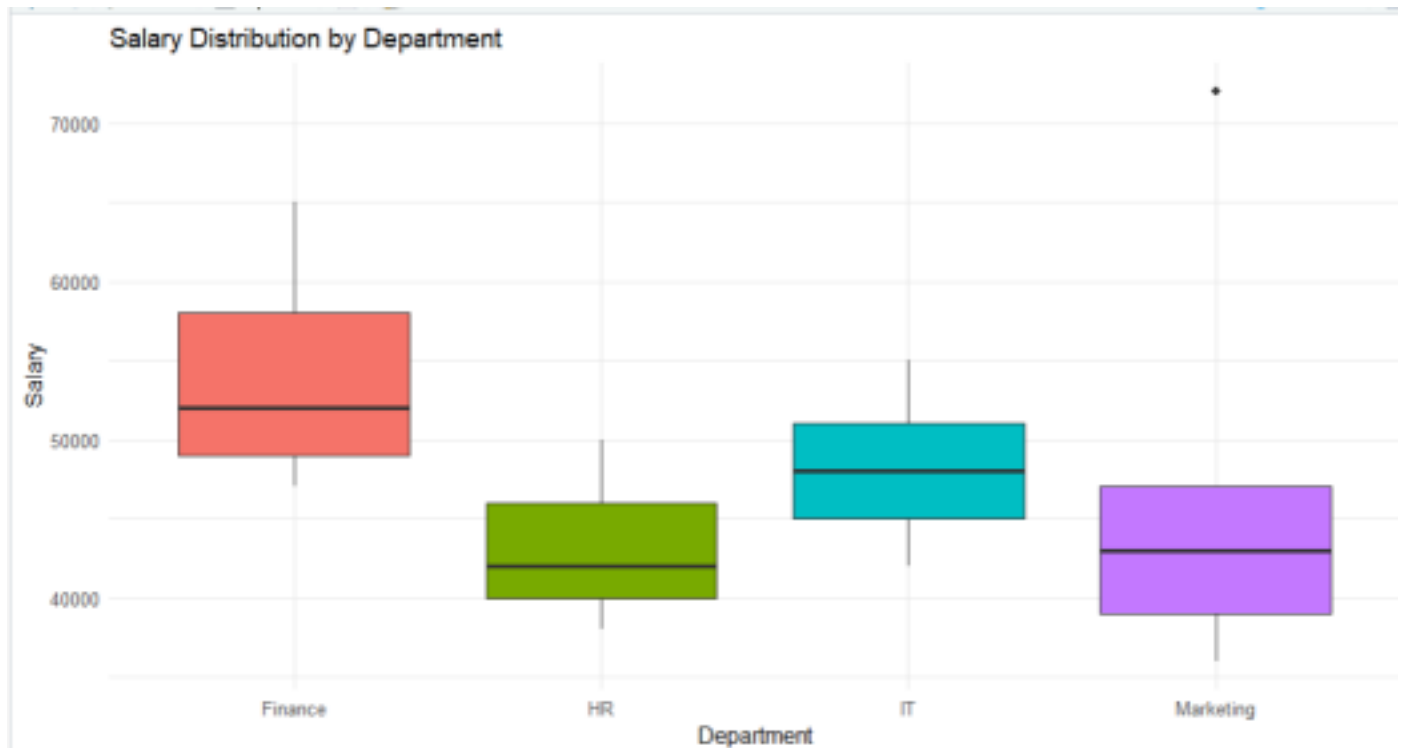
3. Compare the salary distributions of the four departments using suitable distribution plots.

```
qqnorm(employee$Salary,  
  main = "Q-Q Plot of Employee Salaries",  
  xlab = "Theoretical Quantiles",  
  ylab = "Sample Quantiles")
```

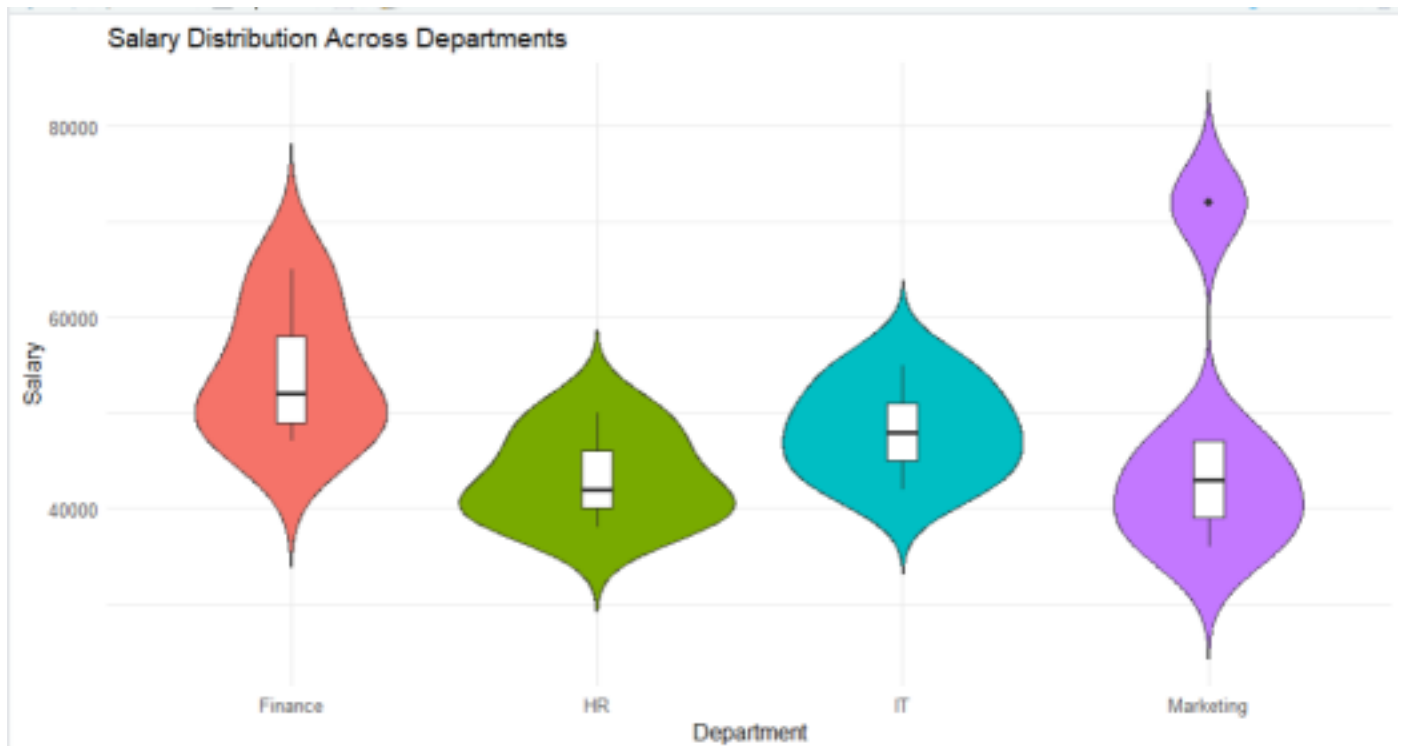
```
qqline(employee$Salary, col = "red", lwd = 2)
```



4. Identify any possible skewness or unusual observations and explain what the plots reveal. `ggplot(employee, aes(x = Department, y = Salary, fill = Department)) +
 geom_boxplot() +
 labs(
 title = "Salary Distribution by Department",
 x = "Department",
 y = "Salary"
) +
 theme_minimal() +
 theme(legend.position = "none")`



```
ggplot(employee, aes(x = Department, y = Salary, fill = Department)) +  
  geom_violin(trim = FALSE) +  
  geom_boxplot(width = 0.1, fill = "white") +  
  labs(  
    title = "Salary Distribution Across Departments",  
    x = "Department",  
    y = "Salary"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```



Dataset:

Employee_ID	Department	Salary
E001	IT	42000
E002	IT	45000
E003	IT	48000
E004	IT	51000
E005	IT	55000
E006	HR	38000
E007	HR	40000
E008	HR	42000
E009	HR	46000
E010	HR	50000
E011	Finance	47000
E012	Finance	49000
E013	Finance	52000
E014	Finance	58000

E015	Finance	65000
E016	Marketing	36000
E017	Marketing	39000
E018	Marketing	43000
E019	Marketing	47000
E020	Marketing	72000

Question 2 – Visualizing Many Distributions Using Bean Plots

Dataset: Student_Performance.csv

Assessment Task

Using the Student_Performance dataset, investigate how scores are distributed across courses. 1. Visualize the score distribution for all four courses at the same time.

```
install.packages("ggplot2") # Run only if not already installed
```

```
library(ggplot2)
```

```
student <- data.frame(
  Student_ID = paste0("S", sprintf("%03d", 1:20)),
  Course = c(
    rep("Python", 5),
    rep("Statistics", 5),
    rep("DataViz", 5),
    rep("MachineLearning", 5)
  ),
  Study_Hours = c(
    5, 7, 9, 11, 14,
    4, 6, 8, 10, 13,
    3, 5, 7, 9, 12,
    6, 8, 10, 13, 16
  ),
  Score = c(
    62, 68, 74, 81, 88,
    58, 64, 71, 79, 86,
```

```

55, 61, 69, 77, 84,
65, 72, 78, 87, 93
)
)

```

student

```

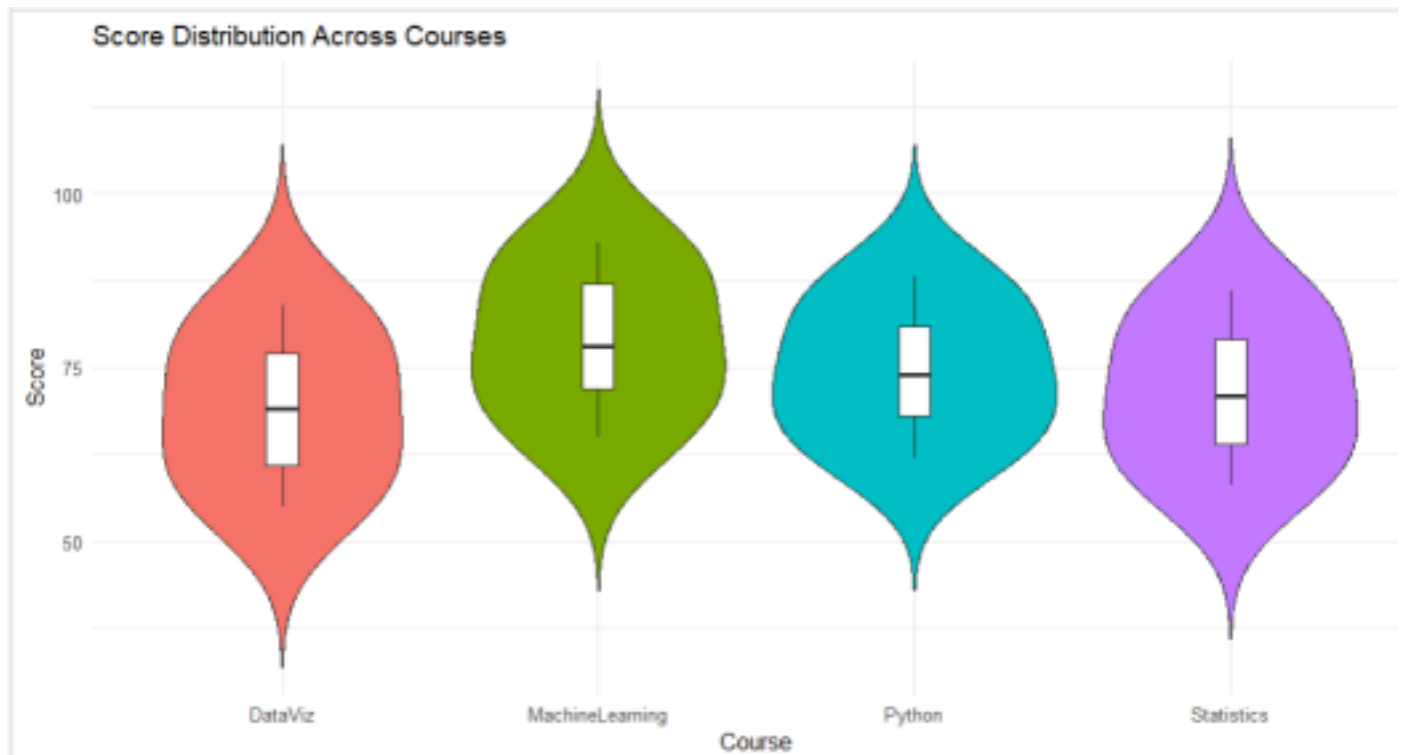
# Student_Score
# A tibble: 20 x 5
#   Student_ID Course Study_Hours Score
#   <int> <chr> <int> <int>
1     S001 Python         5     62
2     S002 Python         7     68
3     S003 Python         9     74
4     S004 Python        11     81
5     S005 Python        14     88
6     S006 Statistics      4     58
7     S007 Statistics      6     64
8     S008 Statistics      8     71
9     S009 Statistics     10     79
10    S010 Statistics     13     86
11    S011 DataViz         3     55
12    S012 DataViz         5     61
13    S013 DataViz         7     69
14    S014 DataViz         9     77
15    S015 DataViz        12     84
16    S016 MachineLearning      6     65
17    S017 MachineLearning      8     72
18    S018 MachineLearning     10     78
19    S019 MachineLearning     13     87
20    S020 MachineLearning     16     93

```

```

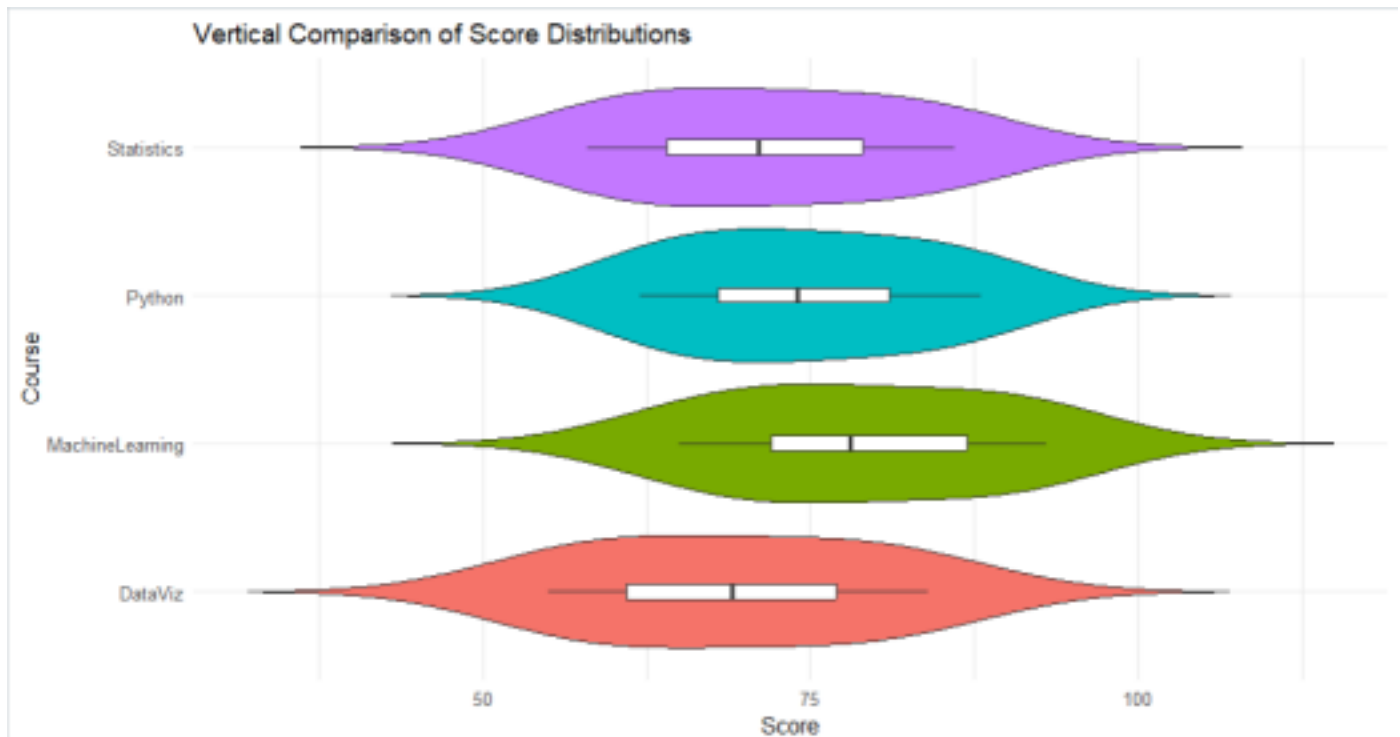
> |
2. Create bean plots (or
an equivalent violin/density-based visualization if a bean-plot package is unavailable). ggplot(student, aes(x =
Course, y = Score, fill = Course)) +
  geom_violin(trim = FALSE) +
  geom_boxplot(width = 0.1, fill = "white") +
  labs(
    title = "Score Distribution Across Courses",
    x = "Course",
    y = "Score"
  ) +
  theme_minimal() +
  theme(legend.position = "none")

```



3. Create a second visualization with the distributions arranged along the vertical axis.

```
ggplot(student, aes(x = Score, y = Course, fill = Course)) +
  geom_violin(trim = FALSE) +
  geom_boxplot(width = 0.1, fill = "white") +
  labs(
    title = "Vertical Comparison of Score Distributions",
    x = "Score",
    y = "Course"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



4. Compare the center, spread, and shape of the distributions and determine which course has the highest and most consistent performance.

```
aggregate(Score ~ Course, data = student,
```

```
  FUN = function(x) c(
```

```
    Mean = mean(x),
```

```
    SD = sd(x),
```

```
    Min = min(x),
```

```
    Max = max(x)
```

```
  ))
```

```
+ 
```

```
))
```

```

  Course Score.Mean Score.SD Score.Min Score.Max
1   DataViz   69.20000  11.71324  55.00000  84.00000
2 MachineLearning  79.00000  11.24722  65.00000  93.00000
3     Python     74.60000  10.28591  62.00000  88.00000
4   Statistics   71.60000  11.23833  58.00000  86.00000
> |
```

Dataset:

Student_ID	Course	Study_Hours	Score
S001	Python	5	62
S002	Python	7	68
S003	Python	9	74
S004	Python	11	81

S005	Python	14	88
S006	Statistics	4	58
S007	Statistics	6	64
S008	Statistics	8	71
S009	Statistics	10	79
S010	Statistics	13	86
S011	DataViz	3	55
S012	DataViz	5	61
S013	DataViz	7	69
S014	DataViz	9	77
S015	DataViz	12	84
S016	MachineLearning	6	65
S017	MachineLearning	8	72
S018	MachineLearning	10	78
S019	MachineLearning	13	87
S020	MachineLearning	16	93

Question 3 – Comparing Proportions with Pie, Side-by-Side and Stacked

Bars Dataset: Customer_Purchases.csv

Assessment Task

Using the Customer_Purchases dataset, analyze purchase proportions.

1. Create a pie chart showing the overall proportion of purchases by Product.

```
install.packages("ggplot2") # Run only if not installed
```

```
library(ggplot2)
```

```
customer <- data.frame(
  Customer_ID = c(
    "C001", "C002", "C003", "C004", "C005", "C006",
    "C007", "C008", "C009", "C010", "C011", "C012"
```

```

),
Region = c(
  "North","North","North",
  "South","South","South",
  "East","East","East",
  "West","West","West"
),
Product = c(
  "Laptop","Phone","Tablet",
  "Laptop","Phone","Tablet",
  "Laptop","Phone","Tablet",
  "Laptop","Phone","Tablet"
),
Purchase_Type = c(
  "Online","Online","Store",
  "Store","Online","Store",
  "Online","Store","Online",
  "Store","Online","Store"
),
Count = c(
  25,35,20,
  30,40,25,
  28,32,18,
  22,38,27
)
)
)

```

Customer

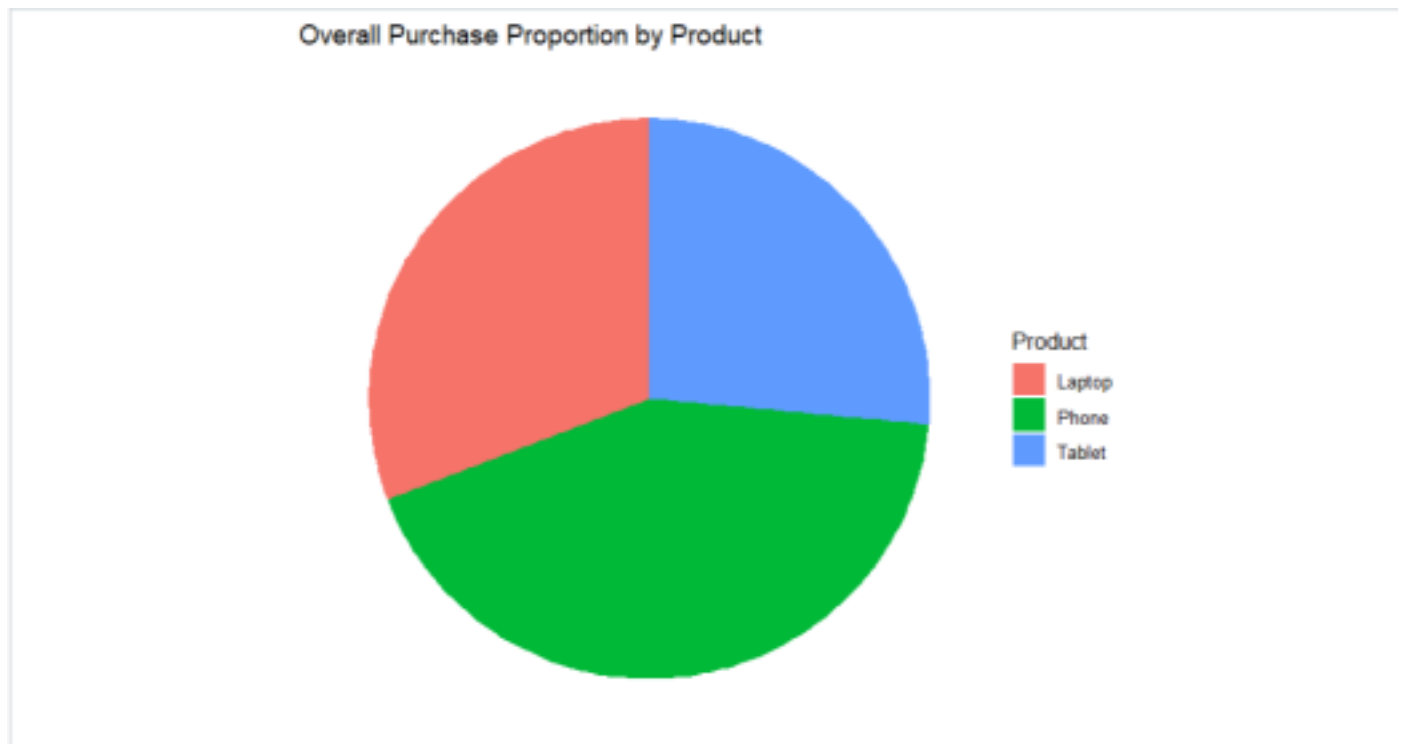
	Customer_ID	Region	Product	Purchase_Type	Count
1	C001	North	Laptop	Online	25
2	C002	North	Phone	Online	35
3	C003	North	Tablet	Store	20
4	C004	South	Laptop	Store	30
5	C005	South	Phone	Online	40
6	C006	South	Tablet	Store	25
7	C007	East	Laptop	Online	28
8	C008	East	Phone	Store	32
9	C009	East	Tablet	Online	18
10	C010	West	Laptop	Store	22
11	C011	West	Phone	Online	38
12	C012	West	Tablet	Store	27

2. Create side-by-side bar charts comparing Product counts across Regions.
 product_total <- aggregate(Count ~ Product, data = customer, sum)

```

product_total
ggplot(product_total, aes(x = "", y = Count, fill = Product)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(
    title = "Overall Purchase Proportion by Product"
  ) +
  theme_void()

```

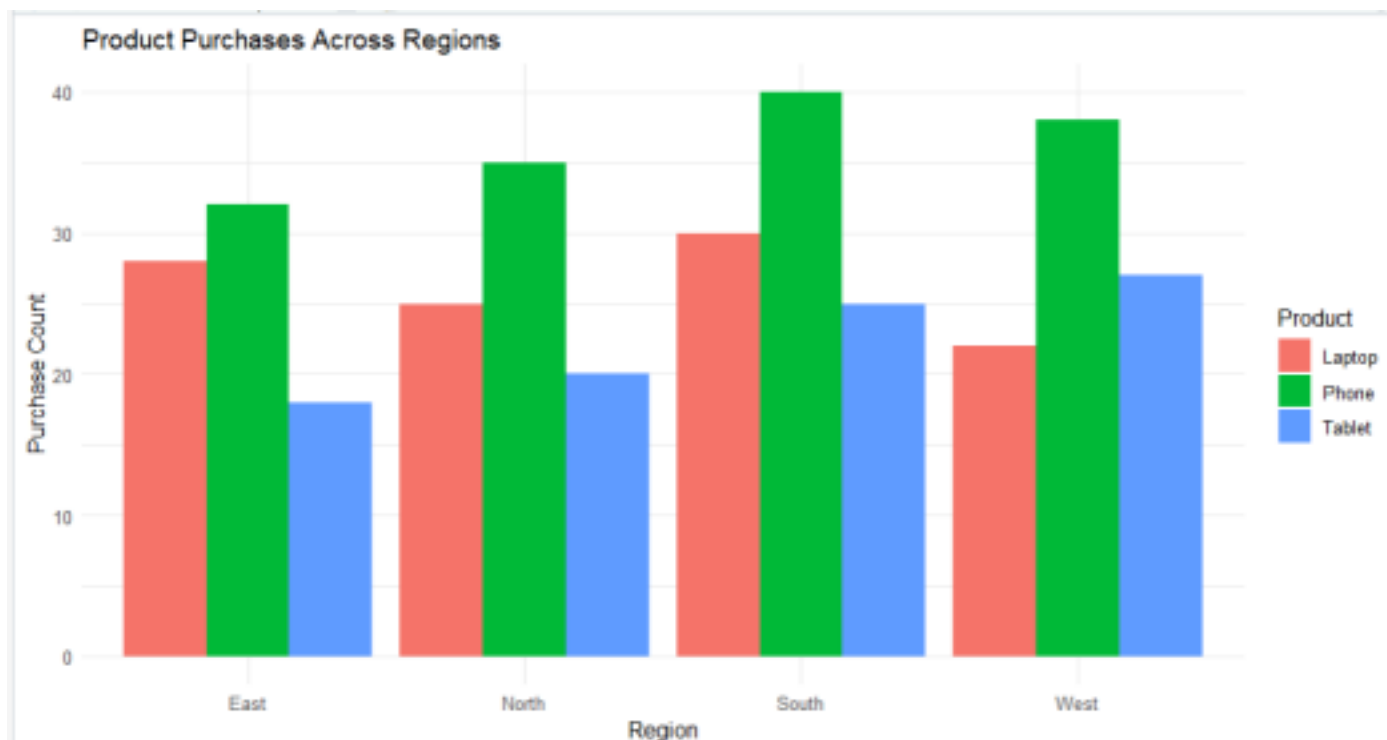


3. Create a stacked bar chart showing product composition within each Region.

```

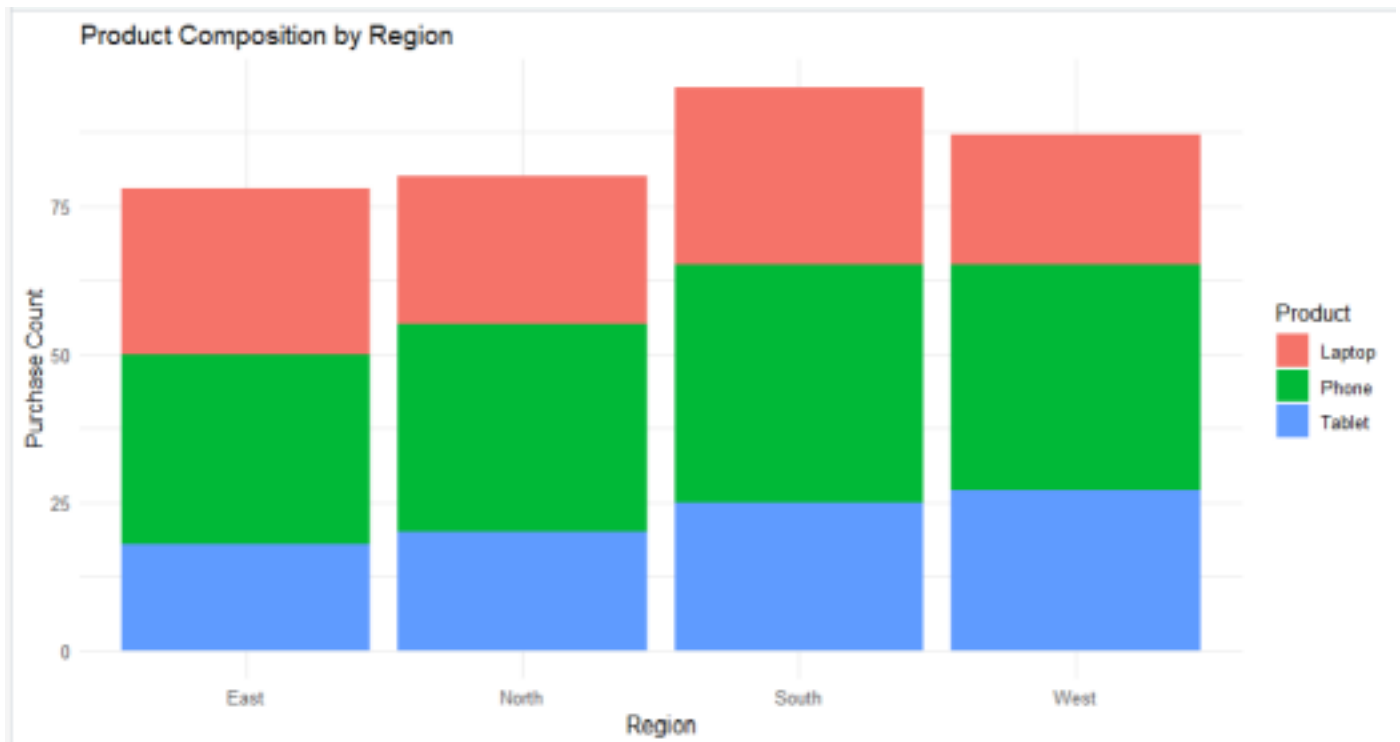
ggplot(customer, aes(x = Region, y = Count, fill = Product)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Product Purchases Across Regions",
    x = "Region",
    y = "Purchase Count"
  ) +
  theme_minimal()

```



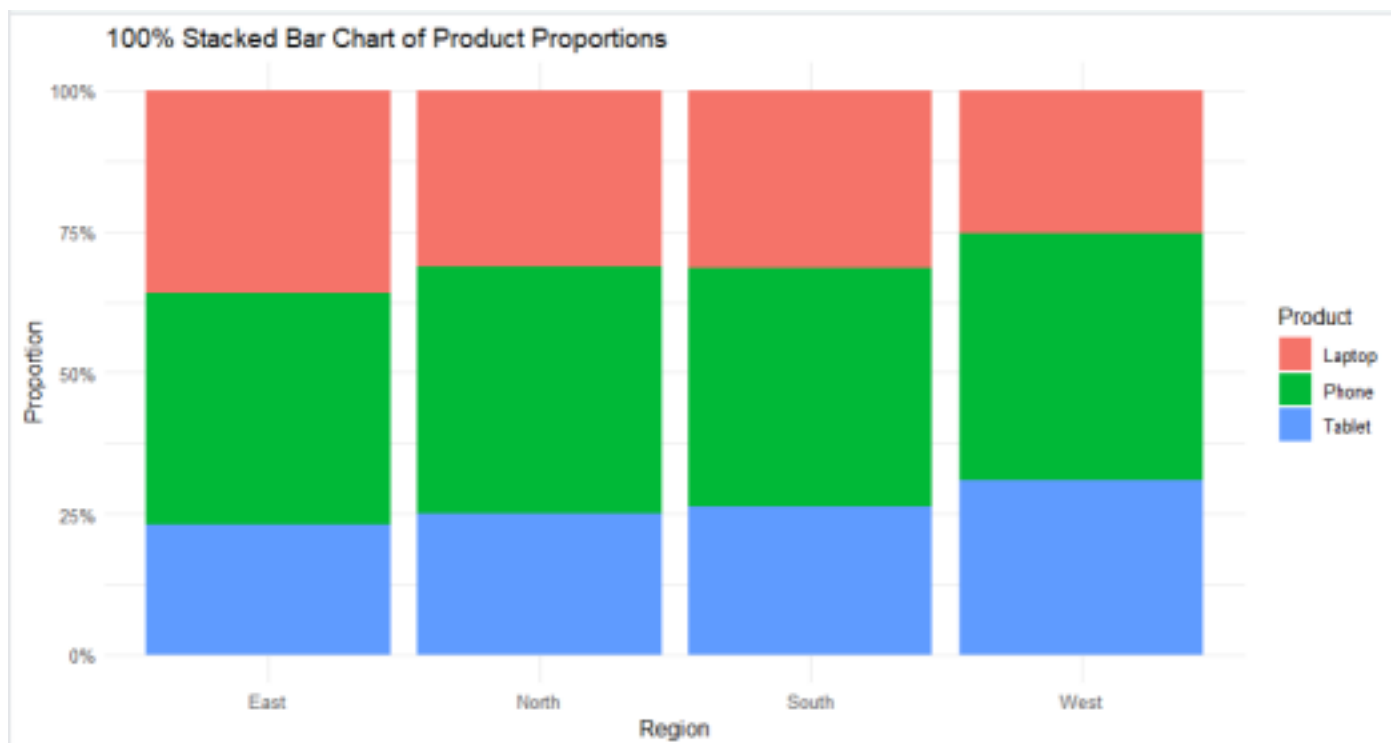
4. Convert the stacked chart into a 100% stacked bar chart and compare proportions rather than raw counts.

```
ggplot(customer, aes(x = Region, y = Count, fill = Product)) +  
  geom_bar(stat = "identity", position = "stack") +  
  labs(  
    title = "Product Composition by Region",  
    x = "Region",  
    y = "Purchase Count"  
  ) +  
  theme_minimal()
```



5. Explain why one of these visualizations may be more appropriate than another for comparing regional proportions.

```
ggplot(customer, aes(x = Region, y = Count, fill = Product)) +
  geom_bar(stat = "identity", position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(
    title = "100% Stacked Bar Chart of Product Proportions",
    x = "Region",
    y = "Proportion"
  ) +
  theme_minimal()
```



Dataset:

Customer_ID	Region	Product	Purchase_Type	Count
C001	North	Laptop	Online	25
C002	North	Phone	Online	35
C003	North	Tablet	Store	20
C004	South	Laptop	Store	30
C005	South	Phone	Online	40
C006	South	Tablet	Store	25
C007	East	Laptop	Online	28
C008		Phone	Store	32
C009		Tablet	Online	18
C010	West	Laptop	Store	22
C011	West	Phone	Online	38
C012	West	Tablet	Store	27

Question 4 – Nested Proportions Using Treemap and Sunburst

Charts Dataset: Company_Sales.csv

Assessment Task

Using the Company_Sales dataset, visualize nested proportions in the sales hierarchy. 1. Create a treemap showing Region → Category → Subcategory, with rectangle size representing Sales.

```
library(treemap)
library(plotly)
library(dplyr)
company <- data.frame(
  Region = c(
    "North", "North", "North", "North",
    "South", "South", "South", "South",
    "East", "East", "East", "East",
    "West", "West", "West", "West"
  ),
  Category = c(
    "Electronics", "Electronics", "Furniture", "Furniture",
    "Electronics", "Electronics", "Furniture", "Furniture",
    "Electronics", "Electronics", "Furniture", "Furniture",
    "Electronics", "Electronics", "Furniture", "Furniture"
  ),
  Subcategory = c(
    "Laptop", "Phone", "Chair", "Desk",
    "Laptop", "Phone", "Chair", "Desk",
    "Laptop", "Phone", "Chair", "Desk",
    "Laptop", "Phone", "Chair", "Desk"
  ),
  Sales = c(
    52000, 38000, 22000, 18000,
    46000, 42000, 25000, 21000,
    58000, 35000, 19000, 23000,
    49000, 44000, 27000, 20000
  )
)

company
```

	Region	Category	Subcategory	Sales
1	North	Electronics	Laptop	52000
2	North	Electronics	Phone	38000
3	North	Furniture	Chair	22000
4	North	Furniture	Desk	18000
5	South	Electronics	Laptop	46000
6	South	Electronics	Phone	42000
7	South	Furniture	Chair	25000
8	South	Furniture	Desk	21000
9	East	Electronics	Laptop	58000
10	East	Electronics	Phone	35000
11	East	Furniture	Chair	19000
12	East	Furniture	Desk	23000
13	West	Electronics	Laptop	49000
14	West	Electronics	Phone	44000
15	West	Furniture	Chair	27000
16	West	Furniture	Desk	20000

2. Create a sunburst chart representing the same hierarchy.

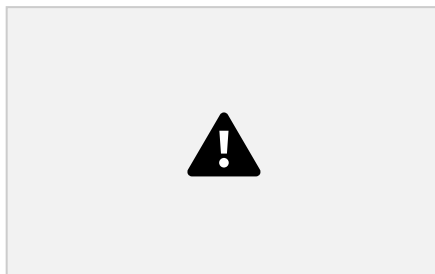
```
treemap(
  company,
  index = c("Region", "Category", "Subcategory"),
  vSize = "Sales",
  type = "index",
  title = "Company Sales Treemap"
)
```



3. Identify the largest region, category, and subcategory by sales.

```
region_sales <- company %>%  
  group_by(Region) %>%  
  summarise(Total_Sales = sum(Sales)) %>%  
  arrange(desc(Total_Sales))
```

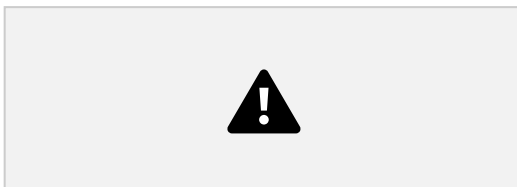
region_sales



4. Compare the readability of the treemap and sunburst chart for identifying nested proportions.

```
category_sales <- company %>%  
  group_by(Category) %>%  
  summarise(Total_Sales = sum(Sales)) %>%  
  arrange(desc(Total_Sales))
```

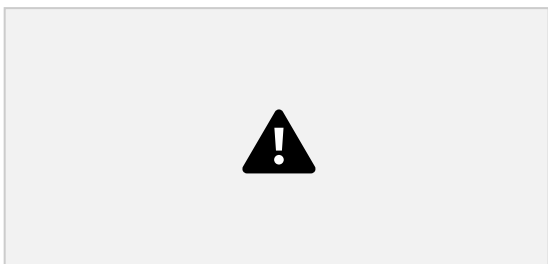
category_sales



5. State one potential limitation of each visualization.

```
subcategory_sales <- company %>%  
  group_by(Subcategory) %>%  
  summarise(Total_Sales = sum(Sales)) %>%  
  arrange(desc(Total_Sales))
```

subcategory_sales



Sunburst Chart

```
sunburst_data <- bind_rows(  
  company %>%  
    group_by(Region) %>%  
    summarise(Sales = sum(Sales)) %>%  
    mutate(  
      labels = Region,  
      parents = ""  
    ) %>%  
    select(labels, parents, Sales),  
  
  company %>%  
    group_by(Region, Category) %>%  
    summarise(Sales = sum(Sales), .groups = "drop") %>%  
    mutate(  
      labels = paste(Region, Category, sep = "-"),  
      parents = Region  
    ) %>%  
    select(labels, parents, Sales),  
  
  company %>%  
    mutate(  
      labels = paste(Region, Category, Subcategory, sep = "-"),  
      parents = paste(Region, Category, sep = "-")  
    ) %>%  
    select(labels, parents, Sales)  
)  
  
fig <- plot_ly(  
  sunburst_data,  
  ids = ~labels,  
  labels = ~labels,  
  parents = ~parents,  
  values = ~Sales,  
  type = "sunburst"  
)  
fig
```



Dataset:

Region	Category	Subcategory	Sales
North	Electronics	Laptop	52000
North	Electronics	Phone	38000
North	Furniture	Chair	22000
North	Furniture	Desk	18000
South	Electronics	Laptop	46000
South	Electronics	Phone	42000
South	Furniture	Chair	25000
South	Furniture	Desk	21000
East	Electronics	Laptop	58000
East	Electronics	Phone	35000
East	Furniture	Chair	19000
East	Furniture	Desk	23000

West	Electronics	Laptop	49000
West	Electronics	Phone	44000
West	Furniture	Chair	27000
West	Furniture	Desk	20000

Question 5 – Flow-Based Proportions with Sankey Diagram and Parallel

Sets Dataset: Student_Career_Path.csv

Assessment Task

Using the Student_Career_Path dataset, analyze how students flow from specialization to internship status and placement sector.

1. Create a Sankey diagram showing Specialization → Internship → Placement_Sector, with flow width based on Students.

```
library(ggplot2)
library(ggalluvial)
library(ggforce)
career <- data.frame(
  Student_Group = c(
    "G1","G2","G3","G4","G5","G6",
    "G7","G8","G9","G10","G11","G12"
  ),
  Specialization = c(
    "Data Analytics","Data Analytics","Data Analytics",
    "AI","AI","AI",
    "Cyber Security","Cyber Security","Cyber Security",
    "Cloud Computing","Cloud Computing","Cloud Computing"
  ),
  Internship = c(
    "Yes","Yes","No",
    "Yes","Yes","No",
    "Yes","Yes","No",
    "Yes","Yes","No"
  ),
  Placement_Sector = c(
    "IT","Finance","IT",
    "IT","Healthcare","IT",
    "IT","Finance","Government",
    "IT","Finance","Government"
  )
)
```

```

),
Students = c(
  18,12,8,
  20,10,6,
  16,9,7,
  15,8,5
)
)

```

```

Career
ggplot(
  career,
  aes(
    axis1 = Specialization,
    axis2 = Internship,
    axis3 =
    Placement_Sector, y =
    Students
  )
) +
  geom_alluvium(
    aes(fill = Internship),
    width = 1/12,
    alpha = 0.8
  ) +
  geom_stratum(
    width = 1/12,
    fill = "grey",
    color = "black"
  ) +
  geom_text(
    stat = "stratum",
    aes(label = after_stat(stratum))
  ) +
  scale_x_discrete(
    limits = c(
      "Specialization",
      "Internship",
      "Placement Sector"
    ),
    expand = c(.1, .1)
  )

```

```

) +
labs(
  title = "Student Career Path Sankey Diagram",
  x = "Career Path",
  y = "Number of Students",
  fill = "Internship"
) +
theme_minimal()

```



2. Create a parallel sets visualization for the same categorical relationships.

```

ggplot(
  career,
  aes(
    axis1 = Specialization,
    axis2 = Internship,
    axis3 = Placement_Sector,
    y = Students
  )
) +
  geom_parallel_sets(
    aes(fill = Internship),
    alpha = 0.6,
    axis.width = 0.15
  ) +
  geom_parallel_sets_axes(
    axis.width = 0.15,
    fill = "grey",
  )

```

```

    color = "black"
  ) +
  geom_text(
    stat = "parallel_sets_axes",
    aes(label = after_stat(axis)),
    hjust = 0.5
  ) +
  scale_x_discrete(
    limits = c(
      "Specialization",
      "Internship",
      "Placement Sector"
    ),
    expand = c(.1, .1)
  ) +
  labs(
    title = "Parallel Sets of Student Career Paths",
    x = "Career Path",
    y = "Number of Students",
    fill = "Internship"
  ) +
  theme_minimal()

```



3. Identify the major flows and compare the placement patterns of students with and without internships.



So

there are 108 students with internships and 26 without internships. **Placement patterns**

For students **with internships**:

- IT = 69
- Finance = 29
- Healthcare = 10

For students **without internships**:

- IT = 14
- Government = 12



4.

Explain which visualization communicates the flow structure more clearly and why.

Sankey diagram

The Sankey diagram communicates the flow structure more clearly.

It makes it easy to follow:

Specialization → Internship → Placement Sector

The width of each connection immediately shows how many students follow that path.

Parallel Sets

The parallel-sets chart is also useful, especially for comparing multiple categorical dimensions simultaneously, but the crossing lines can become harder to follow when there are many categories.

Therefore:

Sankey diagram is better for understanding the overall flow of students through the career pathway.

5. Discuss one way the chart could become misleading if proportions are represented using inappropriate scales, colors, or labels.

The chart can become misleading if the flow widths, colors, or labels do not accurately represent the proportions.

For example, if a flow representing 5 students is displayed almost as wide as a flow representing 20 students, the viewer may incorrectly assume that both flows have similar importance.

Similarly, using confusing colors or misleading percentage labels can make the comparison between internship and non-internship students difficult.

Dataset:

Student_Group	Specialization	Internship	Placement_Sector	Students
G1	Data Analytics	Yes	IT	18
G2	Data Analytics	Yes	Finance	12
G3	Data Analytics	No	IT	8
G4	AI	Yes	IT	20
G5	AI	Yes	Healthcare	10
G6	AI	No	IT	6
G7	Cyber Security	Yes	IT	16
G8	Cyber Security	Yes	Finance	9

G9	Cyber Security	No	Government	7
G10	Cloud Computing	Yes	IT	15
G11	Cloud Computing	Yes	Finance	8
G12	Cloud Computing	No	Government	5

Expected Student Deliverables

- R script (.R) containing data creation/import, data preparation, analysis, and visualization code.
- All required plots with meaningful titles, axis labels, legends, and appropriate scales.
- A short interpretation for each visualization.
- A brief comparison of alternative visualization methods where requested.
- A concluding statement identifying important patterns, differences, or potential misleading visual encodings.

Suggested Assessment Criteria

Criterion	Weight
Dataset preparation	20%
Correct R programming and use of visualization packages	25%
Appropriate choice and construction of visualizations	25%
Interpretation and analytical reasoning	20%
Clarity, labeling, and presentation	10%